



Multiple Integral

Home Assignment-4

Instructions for solving the assignment:

1. Every student is supposed to solve 1 sub-question from each of the main Questions **A, B, C and D** (i.e. total 4 questions) **based on the table given at the end of the assignment.**
2. Write the assignment in the section for Home assignments in your tutorial notebook.
3. Get the assignment checked by your tutorial teacher on or before **15/01/2025**

A) Sketch the following regions

1. The region bounded by $y^2 = 3 - x$ and $y^2 = 2x$
2. The region bounded by $x^2 = 6y$ and $y = x + 2$
3. The region bounded by $x^2 + y^2 = 9$ and $x^2 = 5y$
4. The region bounded by $y \geq 0, x = 6$ and $y^2 = 12 - x$
5. The region bounded by $y = \sqrt{5x - x^2}$ and $y = 0$
6. The region bounded by $x = 1, x = \sqrt{2}, y = 0$ and $\sqrt{2 - x^2}$
7. The region bounded by $y = \sqrt{ax - x^2}, y = \sqrt{a^2 - x^2}, x = 0$ and $x = a$ where a is some constant
8. The region bounded by $y = 2\sqrt{ax}, y = \sqrt{5ax - x^2}, x = 0$ and $x = a$ where a is some constant
9. The region bounded by $x = 0, x = \frac{a}{\sqrt{2}}, y = x$ and $\sqrt{a^2 - x^2}$ where a is some constant
10. The region bounded by $x = 0, x = 1, y = 0$ and $y = \sqrt{x}$
11. The region bounded by $y = 0, y = 1, x = 0$ and $x = \sqrt{2 - y^2}$
12. The region bounded by $x = 0, x = 2, y = \sqrt{2x}$ and $y = 2$

13. The region bounded by $x = 0, x = \frac{a}{2}, y = \frac{x^2}{a}$ and $y = x - \frac{x^2}{a}$ where a is some constant
14. The region bounded by $y^2 + (x - 2)^2 = 1$ and $y^2 + (x - 4)^2 = 4$
15. The region bounded by $y = 0, y = 1, x = 2y, x = 2(1 + \sqrt{1 - y})$

B) Evaluate the following (Use polar/Cylindrical/spherical coordinates if required)

1. $\iint_R (x^2 + y^2) dx dy$ where R is bounded by $y = 4x, x + y = 3, y = 0$ and $y = 2$
2. $\iint_R (x + y) dx dy$ over R is bounded by $x = 0, x = 2, y = x$ and $y = x + 2$
by taking the order first x then y
3. $\iint_R (x + y) dx dy$ Where R is a triangular region with vertices $(0,0), (10,1)$ and $(1,1)$.
4. $\iint_R e^{y^2} dx dy$ Where R is a triangular region with vertices $(0,0), (2,1)$ and $(1,1)$.
5. $\iint_R (x^2 + 2x) dx dy$ Where R region bounded by $y^2 = 3 - x$ and $y^2 = 2x$
6. $\iint_R \frac{xy}{(x+y)^2} dx dy$ where R is the region bounded by $x = 0, x = a, y = \sqrt{ax - x^2}$ and $y = \sqrt{a^2 - x^2}$ where a is the constant
7. $\iint_R \frac{\sqrt{x^2 + y^2}}{y^2} dx dy$ where R is the region bounded by $x = 0, x = a, y = 2\sqrt{ax}$ and $y = \sqrt{5ax - x^2}$ where a is the constant
8. $\iint_R \frac{x}{\sqrt{x^2 + y^2}} dx dy$ where R is the region bounded by $x = 0, x = \frac{a}{\sqrt{2}}, y = x$ and $y = \sqrt{a^2 - x^2}$ where a is the constant
9. $\iint_R \frac{(x^2 + y^2)^2}{x^2 y^2} dx dy$ where R is the region bounded by $x^2 + y^2 - ax = 0$ and $x^2 + y^2 - by = 0$ where a, b are constants
10. $\iint_R y dx dy$ where R is the region in first quadrant bounded by $4x^2 + 9y^2 = 36$

11. $\iiint_R dV$ where R is bounded by $x = 0, x = 1, y = x^2, y = \sqrt{x}, z = 0$ and $z = 2 - x - y$
12. Evaluate $\iiint_R (x^2 + y^2) dV$, where R is the region bounded by the paraboloid $z = x^2 + y^2$ and the plane $z = 4$.
13. $\iiint_R z dV$ where R is the region bounded $x = 0, y = 0, z = 0$ and $x + y + z = 1$
14. $\iiint_R e^{-(x^2+y^2+z^2)} dV$ where R is the sphere of radius 2 defined by $x^2 + y^2 + z^2 \leq 4$
15. $\iiint_R \frac{1}{\sqrt{x^2+y^2+z^2}} dV$ where R is the region bounded cylinder $x^2 + y^2 = 1$ the plane $z = 0$, and the plane $z = 3$.

C) Change the order of the following and hence evaluate (Use polar/Cylindrical/spherical coordinates if required)

- $\int_0^\infty \int_0^x x e^{\frac{-x^2}{y}} dx dy$
- $\int_0^a \int_0^{a-\sqrt{a^2-y^2}} \frac{xy \log(x+a)}{(x-a)^2} dx dy$
- $\int_0^1 \int_x^{\frac{1}{x}} \frac{y}{(1+x)^2(1+y^2)} dx dy$
- $\int_0^1 \int_0^{\sqrt{1-y^2}} \frac{\cos^{-1}x}{\sqrt{1-x^2-y^2}\sqrt{1-x^2}} dx dy$
- $\int_0^1 \int_x^{2-x} \frac{x}{y} dy dx$

6. $\int_0^1 \int_y^1 x^2 e^{xy} dx dy$

7. $\int_0^{\frac{\pi}{2}} \int_0^y \cos 2y \sqrt{1 - a^2 \sin^2 x} dx dy$ where a is constant

8. $\int_0^1 \int_x^{x+1} (xy + y^2) dy dx$

9. $\int_0^1 \int_y^{\sqrt{y}} \sin(x^2 + y^2) dx dy$

10. $\int_0^1 \int_{x^2}^x \sin(x^2 + y^2) dx dy$

D) Solve the following (Use double integral to evaluate area and triple integral to evaluate volume) (Use polar/Cylindrical/spherical coordinates if required)

1. Find the area by using double integral bounded by the $4x^2 + 9y^2 = 36$ in first two quadrants
2. Find the volume of the tetrahedron bounded by the planes $x = 0, y = 0, z = 0$ and $x + y + z = 1$
3. Find the volume of a cylinder of radius R and height h . (Hint: use cylindrical coordinates)
4. Find the volume of the region bounded by the sphere $x^2 + y^2 + z^2 = a^2$ and $z = \sqrt{x^2 + y^2}$ (use spherical co-ordinates)
5. Find the area by using double integral enclosed by $y = x^2, y = 4 - x^2$
6. Find the volume of sphere of radius R (use spherical co-ordinates)

7. Find the volume of the region bounded by the paraboloid $z = x^2 + y^2$ and the plane $z = 4$
8. Find the area of a rectangle with vertices $(0,0)$, $(a, 0)$, (a, b) , and $(0, b)$.
9. Find the area of the annulus bounded by the circles $r = a$ and $r = b$ ($a < b$)
10. Find the area of the region between the ellipses $\frac{x^2}{4} + \frac{y^2}{9} = 1$ and $\frac{x^2}{1} + \frac{y^2}{4} =$

Question Distribution roll numberwise

Roll No.	A	B	C	D
1	7	2	6	6
2	12	15	5	1
3	11	7	1	7
4	15	4	4	9
5	4	10	4	3
6	5	10	6	3
7	6	10	7	10
8	13	4	3	10
9	5	3	10	1
10	11	7	3	4
11	15	6	3	3
12	4	3	3	7
13	6	1	9	6
14	5	10	10	1
15	6	2	2	2
16	12	15	7	7
17	8	7	5	5
18	4	4	5	5
19	11	14	4	6
20	2	12	9	3
21	6	9	1	5
22	7	10	7	6
23	14	7	9	10
24	3	12	5	7
25	4	11	7	10
26	12	15	7	9
27	2	4	3	6
28	15	5	3	4
29	10	6	2	8
30	4	13	7	4
31	15	5	7	10
32	1	11	2	6
33	5	15	10	1
34	2	4	10	8
35	4	6	4	4
36	10	5	1	7
37	6	6	6	8
38	12	12	1	5
39	8	8	9	2
40	4	4	10	3

Roll No.	A	B	C	D
41	9	7	5	6
42	12	14	10	8
43	4	3	4	10
44	11	4	7	2
45	5	12	7	9
46	2	2	8	9
47	10	15	6	3
48	2	10	5	10
49	9	4	9	7
50	13	15	5	1
51	8	1	10	7
52	2	5	10	1
53	3	7	4	9
54	4	14	1	2
55	5	7	6	9
56	2	12	7	6
57	14	11	2	1
58	12	15	2	4
59	9	4	7	7
60	10	5	10	2
61	2	6	4	8
62	10	13	2	5
63	12	5	5	4
64	6	11	2	10
65	5	15	2	3
66	14	4	7	3
67	2	6	10	3
68	15	5	8	1
69	7	6	8	7
70	4	12	2	2
71	10	8	7	3
72	10	4	1	5
73	10	11	5	9
74	4	2	2	8
75	3	4	9	2
76	7	15	4	10
77	6	1	8	7
78	3	5	6	1
79	1	2	1	8
80	10	4	8	2